FOOD DELIVERY TIME PREDICTION

Using Logistic Regression and Convolutional Neural Network

Project Report

Submitted by

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Abstract

Food delivery time is affected by several factors such as distance, traffic, weather, order priority and delivery person experience. In this project, machine learning is used to classify a delivery as Fast or Delayed. The supplied dataset contains 200 delivery records and 15 original columns. Since there is no Fast/Delayed label in the dataset, the median delivery time (72.78 minutes) is used as the classification threshold. Logistic Regression is used as a baseline model and a Convolutional Neural Network (CNN) is developed using a simple image representation of the delivery route. The models are evaluated using standard classification metrics.

1. Introduction

Food delivery services depend on more than the distance between a restaurant and a customer. Traffic, weather, order priority, vehicle type and delivery person experience can affect delivery time. This project studies these factors and uses machine learning to classify an order as Fast or Delayed.

2. Problem Statement

To develop a machine learning system that classifies food deliveries as Fast or Delayed using information such as distance, traffic, weather, order details, ratings and location data.

3. Objectives

• Understand and clean the food delivery dataset.

• Create a Fast/Delayed target from delivery time.

• Perform basic feature engineering and exploratory analysis.

• Build a Logistic Regression model.

• Create a route-image representation for CNN processing.

• Build and train a CNN model.

• Evaluate both models using classification metrics.

• Perform five-fold cross-validation.

• Compare the two approaches.

4. Dataset Description

The dataset contains 200 records and 15 original columns.
Important fields include Order_ID, Customer_Location, Restaurant_Location, Delivery_Time, Distance, Weather_Conditions, Traffic_Conditions,

Delivery_Person_Experience, Order_Priority, Order_Cost, Tip_Amount, Vehicle_Type, Restaurant_Rating, Customer_Rating and Order_Time.
The median Delivery_Time is 72.78 minutes.

5. Methodology

The project was completed by loading and inspecting the data, checking missing values and duplicates, parsing location coordinates, creating the target, preparing the tabular data, training Logistic Regression, creating route images, training the CNN, evaluating both models and performing validation.

6. Data Preprocessing

The dataset was checked for missing values, duplicate rows and data types.
The available data was complete, so no rows needed to be removed because of missing values. Customer and restaurant coordinates were separated into latitude and longitude values for the route representation.

6.1 Target Creation

The original dataset does not contain a Fast/Delayed column. The median Delivery_Time was used as the cutoff. Deliveries at or below 72.78 minutes are labelled Fast and deliveries above it are labelled Delayed.

6.2 Feature Engineering

Order_Hour and Rush_Hour were created from the order-time categories. Traffic and weather were also converted into simple numeric levels for analysis. Delivery_Time was not used as an input feature because it is used to create the target.

7. Exploratory Data Analysis

The notebook contains plots for the Fast and Delayed distribution, distance versus delivery time, traffic conditions, weather conditions and a correlation heatmap. These plots help understand the data before model training.

8. Logistic Regression Model

Logistic Regression was selected as the baseline because the problem has two classes. Numerical features were standardized and categorical features were one-hot encoded. An 80:20 stratified train-test split was used.

8.1 Evaluation Metrics

Accuracy, precision, recall, F1-score and ROC-AUC are used to evaluate the models. A confusion matrix and ROC curve are also included in the notebook.

8.2 Logistic Regression Results

The numerical results should be copied from the executed notebook. This avoids adding values that have not actually been calculated.

9. Convolutional Neural Network

The CNN uses a simple 64 × 64 image representation of the customer and restaurant locations. A line between the two points represents the delivery route. This is a simplified representation and is not a real road map.

9.1 CNN Architecture

The CNN contains convolution, max-pooling, flatten, dense and dropout layers, followed by a sigmoid output. Adam is used as the optimizer and binary cross-entropy is used as the loss function. Early stopping is included during training.

9.2 CNN Results

CNN accuracy, precision, recall, F1-score and ROC-AUC should be taken from the executed notebook. The CNN is not assumed to perform better before testing.

10. Model Validation

Five-fold stratified cross-validation is applied to the Logistic Regression pipeline. A small CNN parameter experiment is also included. The tuning is kept limited because the dataset contains only 200 records.

11. Model Comparison

The final comparison should use the actual results from the notebook. Accuracy, precision, recall, F1-score and ROC-AUC can be compared for both models.

12. Limitations

• The dataset is relatively small for deep learning.

• The Fast/Delayed target is derived from Delivery_Time.

• The CNN route image is a simplified coordinate-based representation.

• Real traffic speed and road information are not directly available.

• Results may change with a larger dataset.

13. Future Scope

A larger real-world dataset can be used in future work. Actual road-network information, live traffic and weather data could be added. Other models such as Random Forest, XGBoost and SVM could also be tested. The model could eventually be connected to a simple web application.

14. Conclusion

This project demonstrates a complete machine learning workflow for food delivery classification. Logistic Regression was used as a baseline and a CNN was developed using a route-image representation. The models can be compared using the evaluation metrics generated by the notebook. Because the dataset is small and the target is derived from delivery time, the results should be treated as a project-level

experiment rather than a production system.

15. References

• Python Documentation – https://docs.python.org/

• Pandas Documentation – https://pandas.pydata.org/docs/

• Scikit-learn Documentation – https://scikit-learn.org/stable/

• TensorFlow Documentation – https://www.tensorflow.org/

• Matplotlib Documentation – https://matplotlib.org/

• Seaborn Documentation – https://seaborn.pydata.org/

Appendix: Figures from the Notebook

The final report can include the figures from the executed notebook: Fast vs Delayed distribution, Distance vs Delivery Time, traffic and weather plots, correlation heatmap, example CNN route image, training curves, confusion matrices, ROC curves and the model comparison chart.